

# SuperDoc math: m:box / m:borderBox

## spec coverage test

Each scenario below tests a specific branch of ECMA-376 §22.1.2.11-14 and §22.9.2.7. Labels describe the expected MathML.

### A. m:borderBox defaults

1. No borderBoxPr (expect notation=box)

2. Empty borderBoxPr (expect notation=box)

### B. ST OnOff value variants (on m:hideTop)

3. m:val=1 (expect top hidden)

4. m:val=true (expect top hidden)

5. m:val=on (Annex L.6.1.3 uses this — expect top hidden)

6. Bare tag hideTop/ (expect top hidden — spec default val=1)

7. m:val=0 (expect NOT hidden)

8. m:val=false (expect NOT hidden)

### C. Individual hide flags in isolation

9. hideBot only (expect top left right)

10. hideLeft only (expect top bottom right)

11. hideRight only (expect top bottom left)

12. hideTop + hideRight — spec §22.1.2.12 example (expect bottom left)

13. All four hides, no strikes (expect fallback to mrow, no border)

## D. Strike directions (all borders hidden)

14. strikeBLTR alone (BL to TR = slash = updiagonalstrike)

15. strikeTLBR alone (TL to BR = backslash = downdiagonalstrike)

16. strikeH alone (horizontalstrike)



17. strikeV alone (verticalstrike)

## E. Strike combinations

18. strikeBLTR + strikeTLBR = X pattern

19. All four strikes



20. Default box + strikeH (expect: box horizontalstrike)



21. hideTop + strikeH (expect: bottom left right horizontalstrike)



## F. Spec Annex L.6.1.3 example

22. ~~a squared = b squared + c squared~~ crossed out with TL-BR diagonal, using m:val=on from spec



## G. m:box scenarios

23. Plain m:box with content == (expect mrow)

24. m:box with boxPr/opEmu — spec §22.1.2.13 example (opEmu currently dropped)

25. m:box with boxPr/noBreak (noBreak currently dropped)

26. m:box with all boxPr flags (all currently dropped)

27. Empty m:box (expect null/dropped)

## H. Nested structures

28. m:box inside m:borderBox

29. m:borderBox inside m:box

30. Nested m:borderBox inside m:borderBox (outer partial-hide + inner strike)

 ~~startfirstmiddlesecondend~~